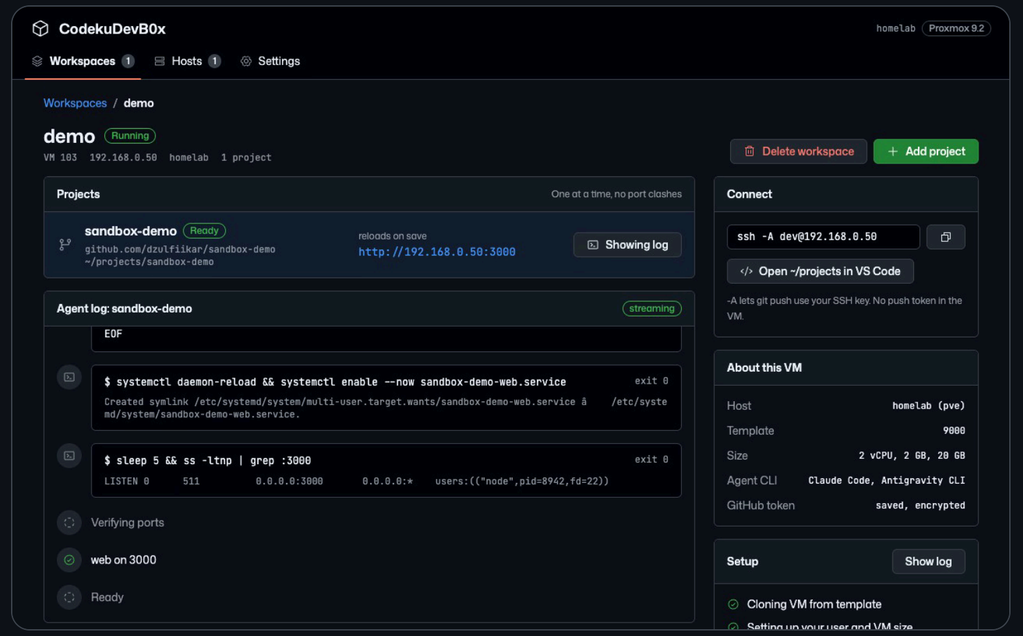
CodekuDevB0x

Run generated code in sandbox VMs on an old laptop or PC running Proxmox, away from your main device. Gemma 4 sets up each repo for you.

Hacktoberfest Hack Day Kota Batam · 31 Oct 2026

`github.com/dzulfiikar/codekudevb0x`



The problem

In the agentic world, code nobody has read runs on your main laptop

01

Code is easy to generate

Agents write code and pull in packages faster than anyone reviews them. Malware can hide in there and you won't notice.

02

It runs next to your data

`npm install`, install scripts and coding agents run with your files, SSH keys, tokens and browser sessions.

03

Isolating it is manual work

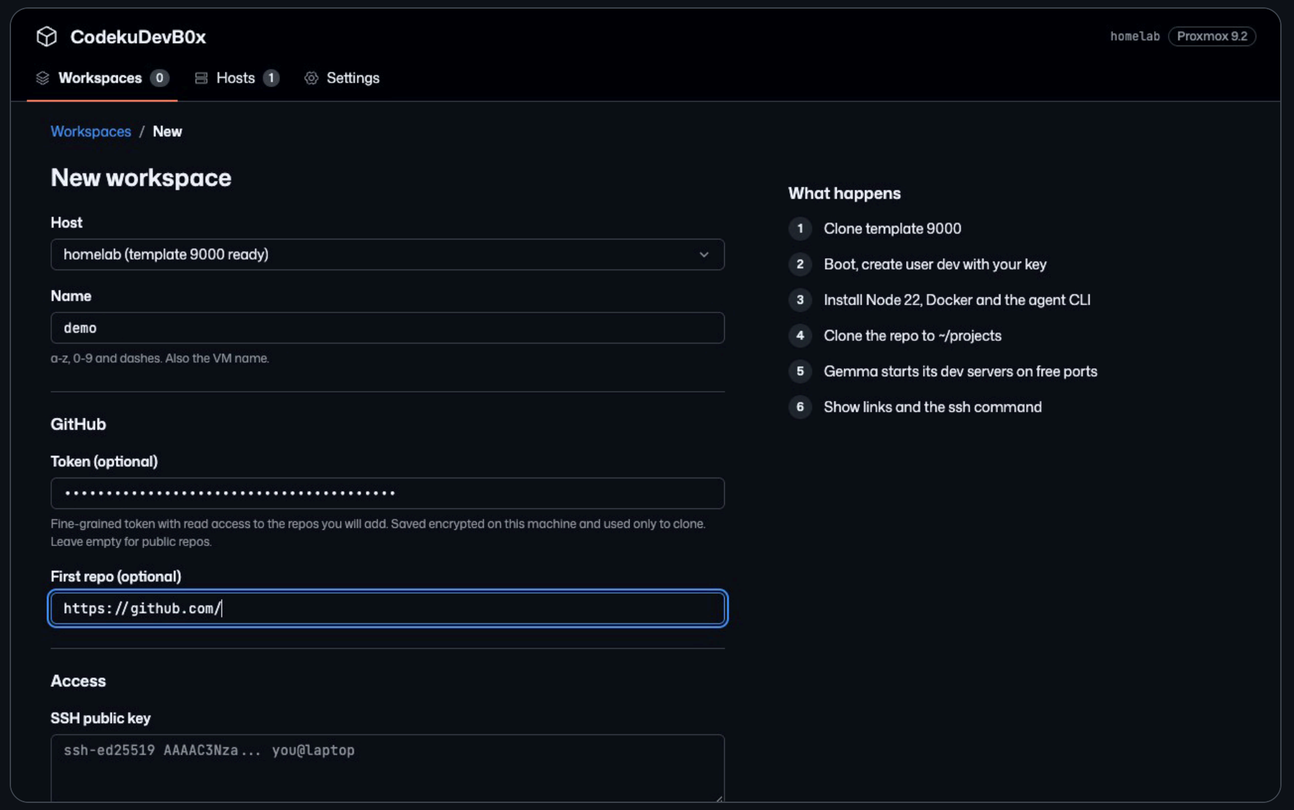
A VM per project means installing tools and getting the repo running again, so most people skip it.

The solution

Sandboxed workspaces, off your main device

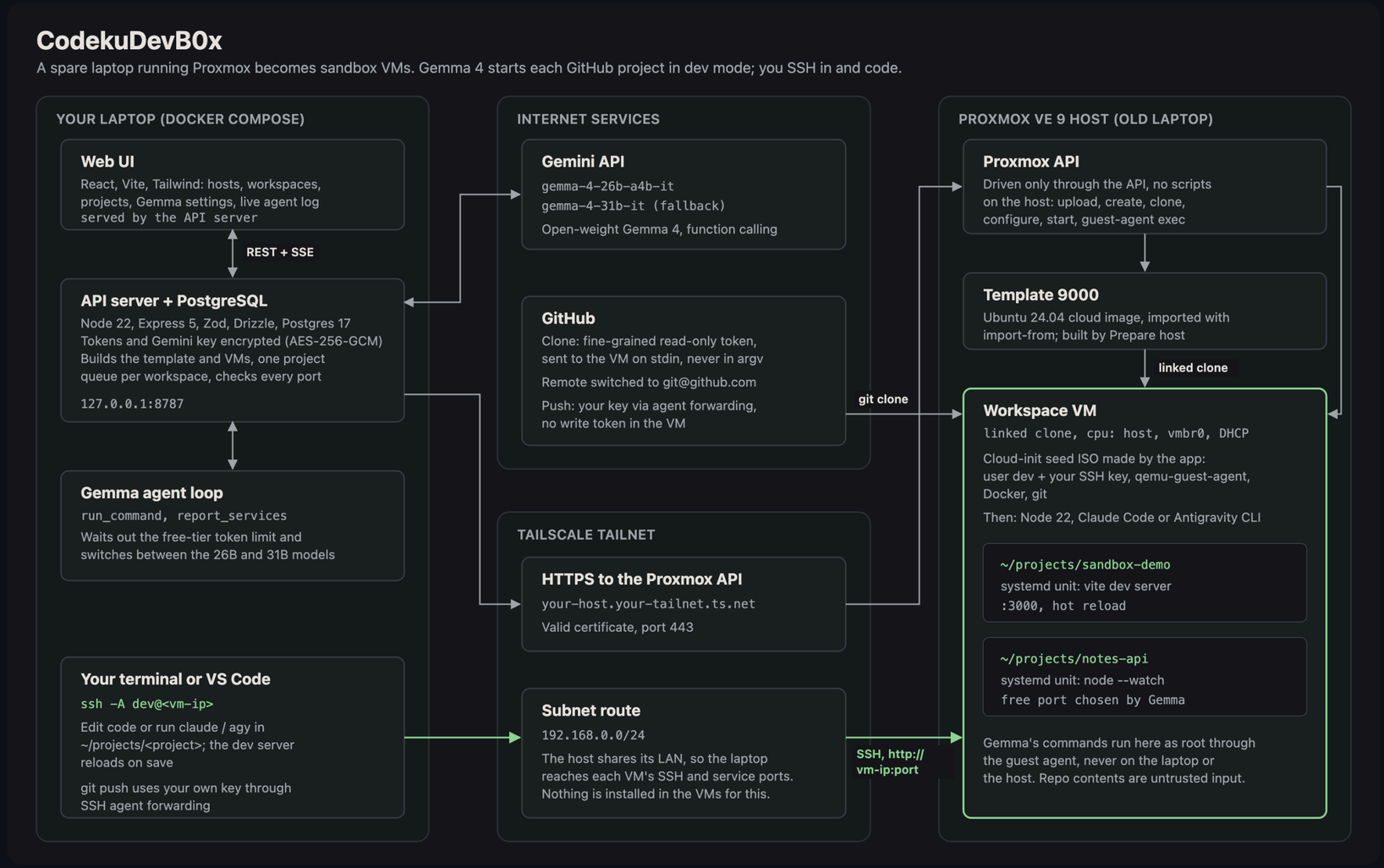
Each workspace is its own VM on your Proxmox box. If a repo carries malware, it gets a VM you can delete, not your laptop.

1. Add your Proxmox host. The app checks it is ready.
2. Create a workspace: GitHub repos, your SSH key, Claude Code or Antigravity CLI.
3. The VM boots. Gemma 4 starts each repo's dev server on a free port.
4. SSH in, edit or run an agent CLI. The page reloads on save.



How it works

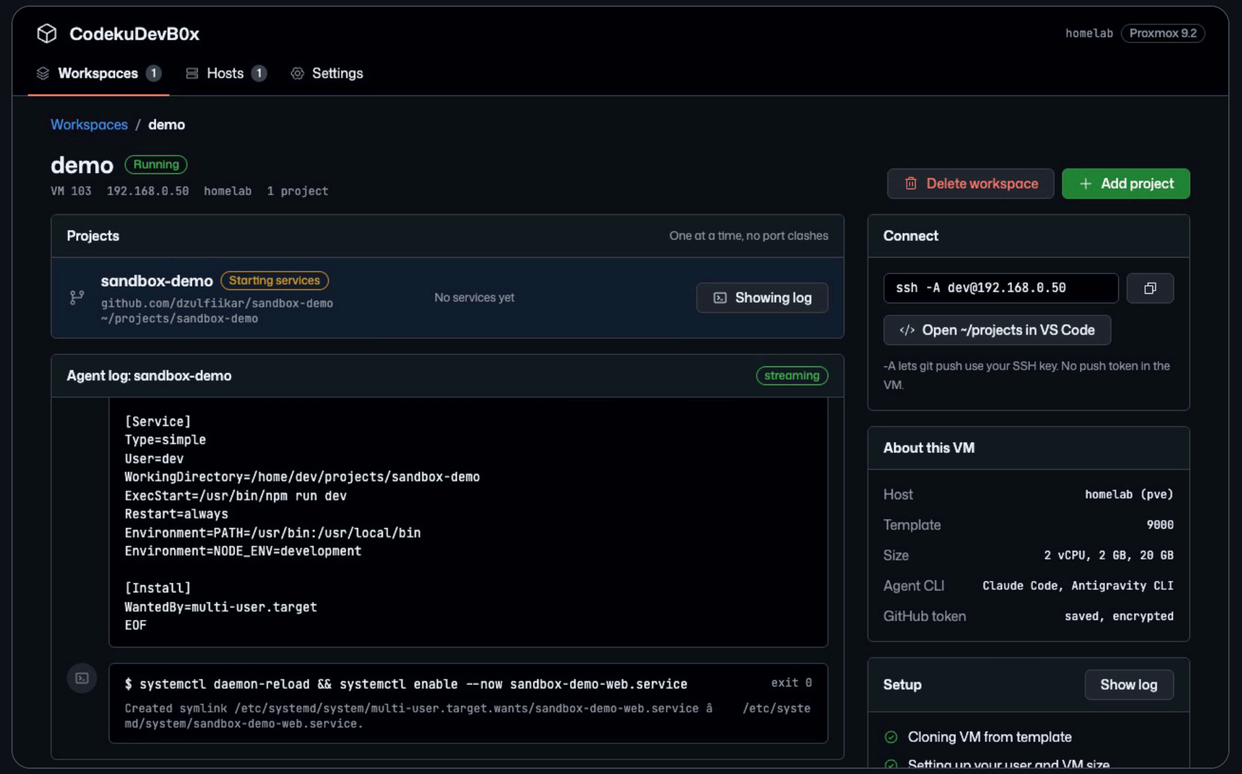
Everything goes through the Proxmox API



No scripts run on the Proxmox host. The app builds its own Ubuntu template through the API.

Gemma 4 is the agent that sets up each repo

- Two tools: `run_command` and `report_services`.
- Works inside the VM through the guest agent: reads the repo, installs dependencies, writes a systemd unit for the dev server.
- Picks ports other projects in the VM don't use. The app checks each port and gives Gemma one more round if one fails.
- On the free tier it waits out the token limit and switches between `gemma-4-26b-a4b-it` and `gemma-4-31b-it`.



Tested on real hardware

End to end on a home Proxmox VE 9.2 server

170 s

Clone the template, boot, install Node, Docker and Claude Code

50 s

Gemma 4 sets up a Vite repo and its dev server answers

39 ms

An edit inside the VM shows up on the page

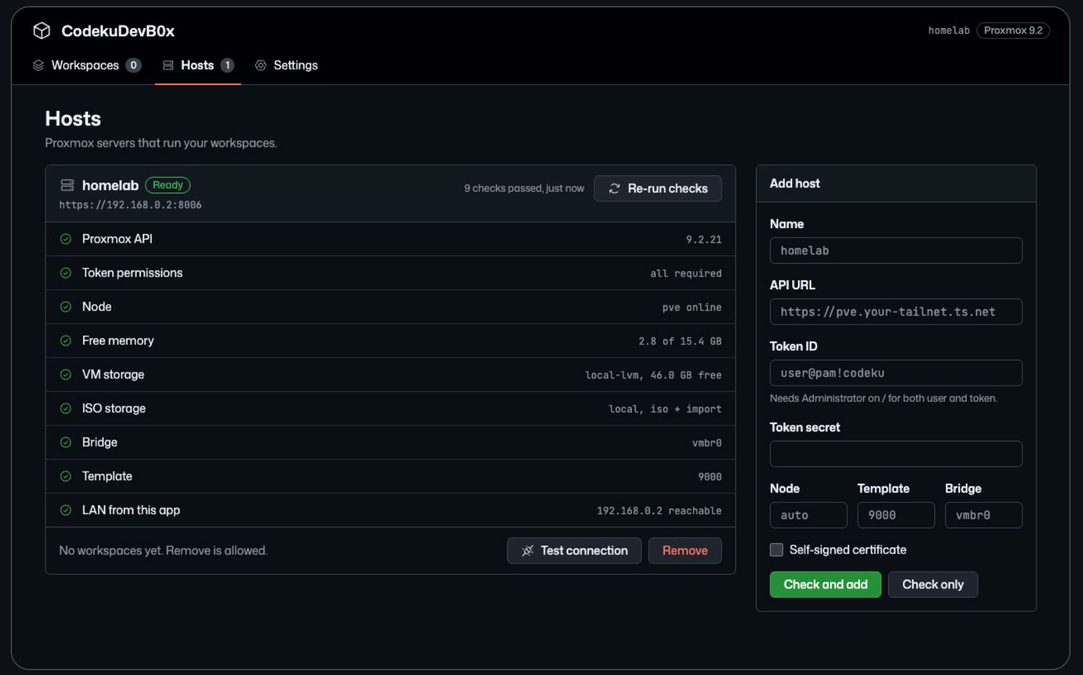
From the automated E2E test (`npm run e2e`): add host, create workspace, wait for the project, call its URLs, edit a file over the guest agent, check the page, delete everything. 8 of 8 steps passed.

Safe by design

What malware in a workspace can't reach

- Your laptop's files, keys and browser sessions. Code, installs, agent CLIs and Gemma's commands all run inside the VM.
- Your other workspaces. Each one is a separate VM.
- Your private SSH key. It stays on the laptop; `git push` borrows it over `ssh -A` only while you're connected.
- Your secrets in the app. Proxmox, GitHub and Gemini keys are encrypted at rest and never sent to the browser.

Done with a repo, or worried about it? Delete the workspace and the VM is gone.



Built with

Open source, MIT licensed, runs on hardware you own

Web	Server	AI	Infra
React 19	Express 5	Gemma 4	Proxmox VE 9.2 API
Vite	Zod	Gemini API	cloud-init
Tailwind CSS v4	Drizzle	Function calling	Tailscale
Primer tokens	PostgreSQL 17		Docker Compose

Built during the Hack Day with Claude Code. The UI was designed first in pen.dev.

Limits and next steps

What it doesn't do yet

Limits today

- Workspace VMs share your home network.
- No login. Run it on your own machine or tailnet.
- Tested on one Proxmox host.
- Projects in a workspace are set up one at a time.

Next

- Put each workspace on its own VLAN behind a firewall.
- Add a login so a small team can share one host.
- Set up several projects at once.
- Test on more hosts and CPU types.

Try it on your own Proxmox

Code	`github.com/dzulfiikar/codekudevb0x`
Demo video	`docs/demo.mp4 in the repo`
Run it	`docker compose up -d`